

Full Process, Sourcing & Test Recipe

India-made kraft fibre to match the US import — Gujarat-local sources, exact proportions to test, and the pass/fail tests. Final direction: recovered kraft sacks + cotton + chemistry (bamboo dropped: NE-grown, needs pulping).

Prepared 2 Oct 2026 · Companion to the research dossier & mill-trial protocol · Prices indicative (India 2025) — confirm by quotation. Proportions are starting points to validate on trial.

CORE IDEA

Used **kraft sacks** (cement/chemical/food) are made from **virgin long-fibre sack kraft** — the same fibre class as imported DLK/DSOCC, but collected in Gujarat. Grade it to a spec, blend with a little cotton waste + a cheap OCC/bagasse base, add strength chemistry/enzyme, and the mill drops it into its existing pulper where the US waste went — **no machine change**.

1 · The process, end to end

- 1
- Collect (Gujarat-local). Used cement/chemical/food **kraft sacks**, box-plant **trim/DLK**, **cotton** mill waste/linter, domestic **OCC**, **bagasse**. All sourced within the state → near-zero freight, daily price, no shipment lock.
- 2
- Sort & grade to a written spec.** Remove plastic liners, strings, tape; separate by fibre quality & cleanliness; control moisture. Define grades (e.g. "India-DLK Grade A / B") the way imported bales carry grades.
- 3
- QC each incoming lot.** Moisture, visible contamination, and a quick handsheet for BF/RCT contribution. Reject or down-grade off-spec lots. **This grading + QC is the value-add that makes it "import level."**
- 4
- Blend to recipe** (section 3). Bale the graded blend as a single stock-keeping product per target BF grade.
- 5
- At the mill — no machine change.** Add the blend to the existing **hydrapulper** where imported waste went; dose **starch/GPAM/PVAm at the wet-end** and **enzyme at the refiner**.
- 6
- Run & sample.** Make paper for a controlled window; pull reel samples.
- 7
- Test vs control** (section 4): BF, RCT, Cobb, GSM, colour, smell, runnability. Cost it per tonne.
- 8
- Standardise & repeat.** Lock the winning recipe + grade spec; QC every future lot to it so quality stays at import level forever.

2 · Sourcing — each material, Gujarat source, price

MATERIAL	WHERE / WHO (GUJARAT-FIRST)	INDICATIVE PRICE
Used kraft sacks (cement/chemical/food) — #1 long fibre	Ahmedabad is a sack-kraft hub; cement/chemical plants, scrap & waste-paper dealers	₹18–26/kg
Box-plant trim / DLK — long fibre	Corrugators/box plants across Gujarat (trim contracts)	₹18–24/kg
Cotton waste / linter — strength booster	Ahmedabad / Rajkot spinning mills; linter from Gujarat (India's main linter region)	spinning waste ~₹100/kg → use small % ; cheaper comber/dropping grades lower
Domestic OCC — base	Local traders / kabadi aggregators	₹14–23/kg
Bagasse — cheap filler	Gujarat sugar mills (depithed)	₹1.4–2.2/kg
Cationic starch	Gujarat Ambuja Exports (Ahmedabad) , Sukhjit, Roquette	₹45–70/kg
GPAM / PVAm / dry-strength resin	Vapi/Vadodara resin cluster , Anmol Polymers, UP Alums, Solenis/Kemira/Ecolab/Buckman	quote (specialty)
Refiner enzyme (cellulase)	Maps Enzymes (Ahmedabad — India's largest) , Advanced Enzyme Technologies (Thane), Novonesis	quote; 0.1–0.5 kg/t

Get trial-quantity samples (a few hundred kg of each fibre; trial kits of each chemical) — usually free/cheap for a documented mill trial. One phone round gets the whole set.

3 · Proportions to test (% of oven-dry furnish)

Run the **CONTROL** (mill's current imported recipe) as benchmark, then these India blends. Chemistry doses are on dry fibre; enzyme at the refiner.

COMPONENT	CONTROL	MIX A 18–22 BF	MIX B 24–28 BF	MIX C 28–32 BF	MIX D TUNE
Domestic OCC	per current	80	60	50	
Imported fibre	e.g. 40	0	0	0	0
Used kraft sacks (long fibre)	–	10	25	35	
Cotton waste (booster)	–	–	5	8	
Bagasse (filler)	–	10	10	7	
— Chemistry & enzyme (added, not part of the 100%) —					
Cationic starch	current	1.0%	1.0%	1.2%	
GPAM	–	0.2%	0.25%	0.3%	
PVAm (RCT grades)	–	–	0.3%	0.4%	
Refiner enzyme	–	0.3 kg/t	0.3 kg/t	0.4 kg/t	

Logic: Mix A tests how far **chemistry alone** (minimal long fibre) reaches for low grades; Mix B is the workhorse for 24–28 BF; Mix C is strength-max for 28–32 BF; Mix D locks the cheapest blend that passes. If a target is missed: starch 0.5–1.5% · GPAM 0.1–0.3% · PVAm 0.2–0.5% · enzyme 0.1–0.5 kg/t · kraft-sack share up to 35–40%. **Also run Mix B with and without enzyme** to prove the enzyme's ₹ value.

4 · Tests & pass targets vs control

PROPERTY	METHOD	INSTRUMENT	PASS TARGET
GSM	ISO 536	Balance	Grade ±2%
BF (Burst Factor)	IS 1060 / ISO 2758 / T403	Mullen burst	Within ±1 BF point
RCT (Ring Crush)	ISO 12192 / TAPPI T822	Ring-crush	≥ 95% of control
CMT (if fluting)	ISO 7263 / T809	Crush tester	≥ 95% of control
Cobb (water abs.)	ISO 535	Cobb tester	Within ±10%
Colour / shade	ΔE or daylight light-box	Spectro / light-box	Visually matched
Smell	5-person blind panel, fresh + 24h bagged	—	Neutral / no off-odour
Moisture	ISO 287	Oven / meter	Within spec
Runnability	Drainage °SR/CSF, breaks, speed	Machine DCS	Speed ≥ 95%, no new breaks

5 · Cost & decision

RECORD (PER TONNE PAPER)	CONTROL	MIX A	MIX B	MIX C	MIX D
BF / RCT					
Colour / Smell (Y/N)	ref				
Speed					
Furnish ₹/tonne					
% imports		0	0	0	0
₹ saved/tonne	—				

Decision: the cheapest Mix that PASSES all of BF, RCT, colour, smell and runnability = your product. Standardise its recipe + the grade spec; QC every future lot to hold import-level quality.

Honest notes: Smell and drainage are the real watch-items — keep them as hard pass/fail lines. 18–32 BF is where a graded kraft-sack blend matches import level; 40–45 BF may still need the highest virgin fibre (trial shows how far). All ₹ figures indicative — **confirm by quotation and validate on a live mill before committing capital.**

Sources: Multiwall/sack-kraft hub & prices — indiamart.com, paperindex.com, tradeindia.com · Cotton waste/linter (Gujarat) — indiamart.com, ScienceDirect "cotton spinning mill waste in papermaking" · OCC/bagasse — indiamart.com, Down To Earth · Starch — Gujarat Ambuja Exports · Enzyme — Maps Enzymes (mapsenzymes.com), Advanced Enzyme Technologies · Dry-strength/GPAM — paperindex.com directory · RCT ISO 12192 / TAPPI T822 — zwickroell.com · Ecolab/Nalco & Solenis dry-strength. Full URLs in research/kraft-paper-import-substitution.md (Appendix B).